

Complete Scientific & Algorithmic Logic Specification

Autonomous SAR Dark-Formation Neural Segmentation, 4D Reverse Lagrangian Backtracking, Multi-Plume Coalescence Dynamics, Spatiotemporal AIS Trajectory Forensics & Bayesian Attribution Engine

Classification: Operational Defense / Enforcement	System Core: OceanTrace-Seg v1.4 / Lagrangian 4D
Date of Issue: 08 September 2026	Compliance: MARPOL 73/78 Annex I / UNCLOS Art. 211
Target Domain: Offshore Maritime Basins & EEZ	Simulation Scope: T-360m (Hindcast) to T+1440m (Forecast)

1. End-to-End Operational Pipeline & Logic Flow

The SpillTheory 4D Digital Twin operates as an automated cyber-physical forensic framework designed to solve the inverse maritime dispersion problem: Given an observed satellite synthetic aperture radar (SAR) oil spill at epoch $t = 0$, determine the exact historical discharge origin coordinates, reconstruct the true trajectory of the slick across past time $t < 0$, identify and correlate all commercial vessels transiting through the discharge envelope, calculate deterministic attribution probabilities, and project forward hydrodynamic dispersion toward coastal risk receptors for $t > 0$.

Phase	Component	Mathematical / Algorithmic Core	Operational Output
Phase 1	SAR Neural Segmentation	C-Band / L-Band backscatter damping, U-Net CNN, Otsu thresholding	GeoJSON polygon, Area (km ²), Topology (Single vs Dual)
Phase 2	MetOcean Ingestion	INCOIS Buoys / HYCOM assimilation: $V_{drift} = V_{curr} + 0.03 \cdot V_{wind}$	Time-aligned 2D vector field (leeway & surface current)
Phase 3	4D Reverse Hindcast	Stochastic Lagrangian SDE with Brownian diffusion: $dX = -V dt + \sqrt{(2K)} dW$	Origin centroid [Lat, Lon], R90 envelope, discharge window
Phase 4	Multi-Plume Dynamics	Confluent unmixing / coalescence: Plume 1 (Red) & Plume 2 (Violet)	Independent origins merging into single unified slick at $t=0$
Phase 5	AIS Forensic Ranking	5-Factor weighted spatiotemporal & behavioral anomaly score	Suspect priority scores (%): Ocean Star (78.2%), Gulf Voyager (76.4%)
Phase 6	Forward Dispersion	Eulerian-Lagrangian coastal transport, Fay spreading, ESI vulnerability	Shoreline impact ETA (~23.5h), Tier-1 response actions

2. Phase 1: Satellite SAR Physics & Neural Segmentation

2.1. Physical Bragg Scattering Damping Mechanism

Spaceborne Synthetic Aperture Radar (Sentinel-1 C-Band at 5.405 GHz, $\lambda = 5.6$ cm; ALOS PALSAR L-Band at 1.27 GHz, $\lambda = 23.6$ cm) relies on resonant Bragg backscattering from wind-generated short ocean capillary and gravity-capillary waves. When an illegal petroleum discharge forms a surface microlayer (sheen or emulsified crude), the fluid's viscoelastic surface film increases dynamic surface tension elasticity and dissipates high-frequency turbulent energy, completely suppressing short capillary waves ($\lambda_B = \lambda_{radar} / (2 \sin \theta_{inc})$). As a consequence, radar pulses undergo specular forward reflection away from the satellite, producing a pronounced normalized radar cross-section (NRCS) damping anomaly: $\Delta\sigma^0 = \sigma^0_{slick} - \sigma^0_{ambient\ sea} \approx -6 \text{ to } -12 \text{ dB}$.

2.2. U-Net Neural Segmentation Architecture & Topology Classification

The satellite tile image is radiometrically calibrated to σ^0_{dB} , speckle-filtered using an adaptive Lee filter (kernel 7x7), and processed by SpillTheory-Seg v1.4.2 (an optimized U-Net with ResNet-34 encoder). The model outputs a continuous pixel probability map $P_{spill}(x, y) \in [0, 1]$. The extracted mask undergoes morphological skeletonization and Gaussian Mixture Model (GMM) spatial clustering to classify between **Single Point-Source Spill** ($K=1$, Area $\approx 8.25 \text{ km}^2$) and **Dual Confluent Spill** ($K=2$, Area $\approx 13.48 \text{ km}^2$).

3. Phase 2: MetOcean Hydrodynamics & Physical Drift Modeling

Surface oil transport is governed by the combined superposition of sea-surface currents (Ekman layer + geostrophic flow), direct atmospheric wind shear leeway, and Stokes wave drift. The net instantaneous drift velocity vector $V_{drift}(x, y, t)$ is modeled as:

Governing Drift Equation:

$$V_{drift}(x, y, t) = V_{current}(x, y, t) + \alpha_{wind} \cdot R(\theta_{leeway}) \cdot V_{wind}(x, y, t)$$

Where:

- $V_{current}$: High-resolution surface current velocity vector (0.32 kn @ 146° SE in Mumbai Basin)
- V_{wind} : 10m atmospheric wind velocity vector (13.0 kn @ 291° NW)
- α_{wind} : Empirical wind leeway transfer coefficient ($\alpha = 0.030 \pm 0.004$)
- $R(\theta_{leeway})$: 2D rotation matrix accounting for Coriolis deflection ($\theta \approx 12^\circ$ to the right of wind in Northern Hemisphere)

3.1. Fay's Three-Stage Oil Spreading Kinetics

The physical expansion of the slick area $A(t)$ across elapsed time t is simulated using Fay's radial spreading formulations across three continuous regimes: (1) *Gravity-Inertial Regime* for $t < t_{_1}$; (2) *Gravity-Viscous Regime* for $t_{_1} \leq t < t_{_2}$; and (3) *Surface Tension-Viscous Regime* for $t \geq t_{_2}$. Evaporative loss and emulsification progressively increase kinematic viscosity ν , decelerating spreading while expanding slick perimeter.

4. Phase 3: 4D Reverse Lagrangian Particle Backtracking (Hindcast)

To identify the unobserved spatial origin and time of discharge, the system reverses the arrow of time in the hydrodynamic advection-diffusion transport equation. A stochastic cloud of $N = 350\text{--}500$ Lagrangian numerical particles is seeded uniformly across the detected SAR slick polygon at $t = 0$. Each particle position $\mathbf{X}_i(t)$ is integrated backward in time according to the reverse Stochastic Differential Equation (SDE):

Reverse Stochastic Differential Equation (Lagrangian Backtracking):

$$d\mathbf{X}_i(t) = - [\mathbf{V}_{\text{drift}}(\mathbf{X}_i, t)] dt + \sqrt{(2 \cdot \mathbf{K}_h \cdot dt)} \cdot \boldsymbol{\xi}_i$$

Where:

- dt : Numerical integration time step ($dt = -300$ seconds backward)
- $\mathbf{K}_h$: Horizontal turbulent eddy diffusivity coefficient ($\mathbf{K}_h = 2.5 \text{ m}^2/\text{s}$)
- $\boldsymbol{\xi}_i \sim N(0, \mathbf{I}_2)$: 2D standard Gaussian white-noise vector modeling sub-grid turbulent mixing

4.1. Probable Origin Centroid & R_90 Uncertainty Envelope

For each candidate past discharge hour $\tau \in [1, 12]$ hours, the ensemble centroid and spatial dispersion covariance are computed:

$$\mathbf{X}_{\text{origin}}(\tau) = (1/N) \sum \mathbf{X}_i(-\tau), \quad R_{90}(\tau) = \text{Percentile}_{90}(\| \mathbf{X}_i(-\tau) - \mathbf{X}_{\text{origin}}(\tau) \|)$$

The optimal discharge epoch $\tau^* = 5.2$ hours (-5h, 02:47 UTC) minimizes the particle ensemble variance and maximizes spatial overlap with commercial navigation corridors, defining **Probable Origin #1 at 18.048°N, 72.525°E ($R_{90} = 3.96 \text{ km}$)**.

5. Phase 4: Multi-Plume Coalescence Dynamics (Scenario: 2 Ships Leaking)

When two vessels discharge oily ballast or bilge waste independently in the same offshore basin, the resulting plumes advect along converging metocean vectors and coalesce into a single merged slick. The SpillTheory engine models this multi-source deconvolution forward and backward in time:

5.1. Dual Discharge Initiation at $t = -5\text{h}$ (02:35–02:47 UTC)

- Plume #1 (Primary / Red)**: Initiated as MT OCEAN STAR (Crude Tanker, MMSI 419001284) transits through **Origin #1 (18.048°N, 72.525°E)** at 02:47 UTC heading 125° SE. Discharge volume $\approx 2,200$ bbl, initial plume area $A_{_1}(-5\text{h}) = 0.75 \text{ km}^2$.
- Plume #2 (Secondary / Violet)**: Initiated as GULF VOYAGER (Chemical Tanker, MMSI 419002931) transits through **Origin #2 (18.062°N, 72.540°E)** at 02:35 UTC heading 310° NW. Discharge volume $\approx 1,640$ bbl, initial plume area $A_{_2}(-5\text{h}) = 0.60 \text{ km}^2$.

5.2. Hydrodynamic Advection & Spatial Coalescence ($t = -5\text{h} \rightarrow 0\text{h}$)

As time advances from $t = -300$ min toward $t = 0$ min, both plumes advect under local wind leeway and current. Let $\Omega_{_1}(t)$ and $\Omega_{_2}(t)$ denote the bounding spatial domains of Plume 1 and Plume 2. The coalescence condition is defined by the Hausdorff intersection boundary:

$$\text{dist}(\partial\Omega_{_1}(t), \partial\Omega_{_2}(t)) \leq \epsilon_{\text{coalescence}} \Rightarrow \Omega_{\text{unified}}(t) = \Omega_{_1}(t) \cup \Omega_{_2}(t)$$

Between $t = -60$ min and $t = 0$ min, the plumes deform, overlap, and merge into a single contiguous 13.48 km^2 anomaly detected by Sentinel-1 at 04:32 UTC ($t = 0$).

6. Phase 5: Spatiotemporal AIS Reconstruction & Forensic Attribution

Historical AIS messages within a $50 \text{ km} \times 50 \text{ km}$ bounding box across $t \in [-8\text{h}, +2\text{h}]$ are ingested. Out of 284 candidate merchant vessels, 17 entered the forensic corridor. The attribution engine computes a multi-factor forensic ranking score $S_{\text{attribution}} \in [0, 100]$:

Composite Attribution Score Formulation:

$$S_{\text{attribution}} = w_{_1} \cdot S_{\text{prox}} + w_{_2} \cdot S_{\text{traj}} + w_{_3} \cdot S_{\text{temp}} + w_{_4} \cdot S_{\text{ais}} + w_{_5} \cdot S_{\text{behavior}}$$

Factor Weightings & Mathematical Definitions:

- S_{prox} (Weight $w_{_1} = 0.30$) - Origin Proximity: $S_{\text{prox}} = 100 \cdot \exp(-d_{\text{min}}^2 / (2 \cdot \sigma_r^2))$, where $\sigma_r = 3.5 \text{ km}$
- S_{traj} (Weight $w_{_2} = 0.25$) - Trajectory Vector Collinearity: $S_{\text{traj}} = 100 \cdot \max(0, \cos(\theta_{\text{vessel}} - \theta_{\text{drift}}))$
- S_{temp} (Weight $w_{_3} = 0.20$) - Temporal Discharge Window Alignment: $S_{\text{temp}} = 100 \cdot \exp(-\Delta t^2 / (2 \cdot \sigma_t^2))$, $\sigma_t = 45 \text{ min}$
- S_{ais} (Weight $w_{_4} = 0.15$) - Transponder Continuity & Rate: $S_{\text{ais}} = 100 - \text{Penalties}(\text{AIS Gaps, Spoofed MMSI, Ping Drops})$
- S_{behavior} (Weight $w_{_5} = 0.10$) - Speed Anomaly & Maneuver: $S_{\text{behavior}} = 100$ if $(\Delta \text{SOG} > 5 \text{ kn or } \Delta \text{Course} > 30^\circ \text{ at origin})$ else 50

Table 6.1: Forensic Evidence Matrix for Filtered Suspect Vessels

Rank & Vessel Name	Type & Flag	Origin Distance	Discharge Time	Anomaly Details	Attribution Score
#1 MT OCEAN STAR IMO 9384910	Crude Tanker Liberia (LR)	2.8 km from Origin 1	02:47 UTC (Δt = 0 min)	Speed drop 12.4→5.8 kn; 35° course deviation	78.2% Dual (91.7% Single)
#2 GULF VOYAGER IMO 9412089	Product Tanker Marshall Is. (MH)	3.1 km from Origin 2	02:35 UTC (Δt = 12 min)	Heading shift 310°→335° along Origin 2 corridor	76.4% Dual (Secondary)
#3 MV BLUE HORIZON IMO 9245178	Bulk Carrier Panama (PA)	9.8 km from Origin	03:15 UTC (Δt = 72 min)	Constant speed 12.2 kn; No maneuvering	63.8% (Cleared)
#4 MT ARABIAN PEARL IMO 9553147	Product Tanker Singapore (SG)	14.5 km from Origin	01:15 UTC (Δt = 110 min)	Maintained standard track; Outside error bound	52.1% (Cleared)

7. Phase 6: Forward Drift Forecasting & Shoreline Risk Modeling

From the detection epoch $t = 0h$ forward to $t = +48h$, the unified slick polygon $\Omega(t)$ is advected forward using the continuous Eulerian-Lagrangian trajectory spine. Concurrently, coastal vulnerability is quantified using the Environmental Sensitivity Index (ESI):

7.1. Coastal Vulnerability & Shoreline Impact Formulation

$$\text{Risk Score}_{\text{zone}} = \sum w_j \cdot \text{ESI}_j \cdot \exp(-d_j/\lambda) \cdot (1/\text{ETA}_j)$$

- **Primary Impact Zone:** Murud-Janjira Artisanal Fisheries & Mangrove Reserve (Raigad Coastal Sector, Maharashtra, 18.298°N, 72.962°E).
- **Projected Coastal ETA:** T + 23.5 hours (08 Sep 04:02 UTC).
- **Risk Classification:** HIGH (ESI 8.4 / Protocol Tier-1 NOSDCP Activated).
- **Mitigation Action Plan:** Deployment of 800m offshore heavy-duty containment booms across Rajpuri Creek ingress; dispatch of Indian Coast Guard Interceptor Craft (C-432) with dynamic disc skimmers to the T+12h drift waypoint.

8. Master Parameter & Formulation Reference

Parameter / Symbol	Nominal Value	Physical Unit	Operational Interpretation
Radar Frequency (f_0)	5.405 (C) / 1.27 (L)	GHz	Microwave Bragg scattering resonance carrier frequency
NRCS Damping ($\Delta\sigma_{\text{NRCS}}$)	-9.2 to -12.4	dB	Backscatter attenuation over surface viscoelastic hydrocarbon
Wind Leeway Coeff (α_{wind})	0.030 ± 0.004	dimensionless	Fractional transfer of 10m wind velocity to surface oil film
Coriolis Angle (θ_{leeway})	12.0° Right	degrees	Northern Hemisphere wind leeway deflection angle
Eddy Diffusivity (K_h)	2.5	m² / s	Sub-grid horizontal turbulent dispersion coefficient
Discharge Age (τ^*)	5.2 (Window: 4.8–6.5)	hours	Optimal backward Lagrangian convergence epoch (-5h)
Origin 1 Coordinates	18.048°N, 72.525°E	degrees	Reconstructed discharge centroid for MT Ocean Star (Plume 1)
Origin 2 Coordinates	18.062°N, 72.540°E	degrees	Reconstructed discharge centroid for Gulf Voyager (Plume 2)
Slick Area ($t=0$)	13.48 (Dual) / 8.25 (Single)	km²	Observed polygon surface coverage at Sentinel-1 SAR epoch
Attribution MT Ocean Star	78.2% (Dual) / 91.7% (Single)	percent	Bayesian spatiotemporal & behavioral attribution ranking score
Attribution Gulf Voyager	76.4% (Dual)	percent	Bayesian spatiotemporal attribution for secondary discharge
Shoreline Impact ETA	23.5	hours	Estimated time of arrival to Murud-Janjira coastal aquaculture

Summary & Enforcement Conclusion:

The SpillTheory 4D Digital Twin successfully bridges spaceborne SAR remote sensing with physical oceanography and maritime intelligence. By mathematically unmixing coalescing oil slicks and reconstructing reverse Lagrangian trajectories, the system provides decisive, objective, and verifiable forensic proof to support environmental enforcement authorities in holding polluting vessels accountable under international maritime law.

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